

Wenjuan Lin

📍 Haining, Zhejiang ✉ wenjunl4@illinois.edu 🔗 <https://latentlin2512.github.io/> 🌐 LatentLin2512

Education

University of Illinois Urbana-Champaign

Sep. 2023 – Jun. 2027

Bachelor of Science in Computer Engineering

- GPA: 3.98/4.00

Zhejiang University

Sep. 2023 – Jun. 2027

Bachelor of Engineering in Electrical and Computer Engineering

- GPA: 3.94/4.00, Rank: 2/71

Honors and Awards

National Scholarship

2024

Awarded by the Ministry of Education of China for outstanding academic and overall performance.

Dean's List

2024, 2025

Awarded for outstanding academic performance at the University of Illinois Urbana-Champaign.

Second Prize Scholarship for Academic Excellence

2024, 2025

Awarded by ZJUI; granted to only two students in the major annually.

Meritorious Winner in Mathematical Contest in Modeling

2024

Awarded to teams ranking in the top 9% worldwide for outstanding performance in mathematical modeling.

Second Prize & Best Essay Prize in Concrete Dragon Boat Race

2025

Awarded by Zhejiang University for outstanding performance in the Concrete Dragon Boat Race competition.

Research Experience

Underwater Navigation under Communication and Visibility Constraints

Zhejiang, China

Advisors: Wenguan Wang, Zhejiang University

Jun. 2026 – Present

- Conducting research on object-goal navigation for autonomous underwater agents in communication- and visibility-constrained environments.
- Exploring efficient system designs that better leverage the visual and semantic capabilities of pretrained foundation models to improve navigation performance and generalization.
- Investigating the adaptation of indoor VLN and ObjectNav methodologies to underwater navigation.

Vision-Language-Action Model Exploration

Zhejiang, China

Advisors: Siliang Tang, Guoming Wang, Zhejiang University

Jun. 2025 – Aug. 2025

- Conducted systematic study of recent Vision-Language-Action (VLA) and world models papers.
- Reproduced open-source VLA pipelines under constrained computational resources, building hands-on experience in experiment reproduction, debugging, model training, and performance evaluation.
- Awarded **Outstanding Project** by ZJU-UIUC.

Course Projects

Autonomous Drone Racing | 1st Place in the Drone Racing Competition

Illinois, United States

ECE484 Principles of Safe Autonomy, Instructor: Sayan Mitra

[Project Website](#) 🔗

- Developed a ROS 2-based autonomous drone racing pipeline for Crazyflie 2.1, integrating IMU and motion capture (MoCap) sensors.
- Designed a classical controller (sparse planning + PID control), with systematic parameter tuning, achieving the fastest hardware racing time among all teams.
- Trained a PPO policy using JAX in a drone racing simulator, outperforming the classical controller in simulation and successfully deploying it on simple real-world tracks.

GPT-2 CUDA Kernel Optimizations | Top 3 in Performance Competition

Illinois, United States

ECE408 Applied Parallel Programming, Instructor: Volodymyr Kindratenko

[Project Report](#) 🔗

- Implemented CUDA kernel optimizations including Tensor Cores, FlashAttention, and warp-level reduction

techniques for GPT-2 acceleration.

- Identified compute, memory, and warp-level bottlenecks using NVIDIA Nsight Systems and Nsight Compute.

Land.io FPGA Two-Board Game | Top 3 Final Project

Illinois, United States

ECE385 Digital Systems Laboratory, Instructor: Zuofu Cheng

[Project Report](#)

- Built a real-time two-player FPGA game across synchronized boards using SystemVerilog and MicroBlaze.
- Implemented a master-slave SPI protocol for inter-board synchronization, along with custom AXI4-Lite peripherals, HDMI rendering, and DDR3-backed video/audio playback.

Extracurricular Activities

EKF-Based State Estimation for a Student Rocket

Illinois, United States

Guidance, Navigation & Control Team Member | [Illinois Space Society](#)

Sep. 2025 – Feb. 2026

- Contributed to debugging and validation of a student rocket state-estimation pipeline based on Extended Kalman Filtering, identifying modeling and coordinate-frame inconsistencies.
- Participated in the LUNA 1.0 launch and served as a Launch Operations team member for LUNA 2.0.

Competitions

Second Prize in Sixth College Student Concrete Dragon Boat Race

Zhejiang, China

Bo Yan Team, Advisor: Shilei Zhang

[Project Poster](#)

- Designed and implemented the electrical control system and documented the system design.
- Collaborated with four teammates to design, build, and test a concrete dragon boat.
- Competed in both sprint and obstacle racing events.

Meritorious Winner in Mathematical Contest in Modeling

Zhejiang, China

Teammates: Siyuan Gong, Xinyi Zheng, Advisor: Wee-Liat Ong

[Project Report](#)

- Developed the Sea Lamprey Population Model using Fisher's Principle, the Leslie Matrix, and the Lotka-Volterra Model to simulate the dynamics of sea lamprey populations and their impact on ecosystems.
- Utilized MATLAB to stimulate difference equations and visualize the results of the mathematical model through various plots.
- Wrote the report presenting the final results, addressing how sex ratio variation influences both sea lamprey populations and the surrounding ecosystem.

Leadership & Service

Main Manager in Publicity Department

Zhejiang, China

ZJU-UIUC Institute

Jul. 2024 – Jun. 2025

- Managed the student publicity team, overseeing recruitment, interviews, and regular meetings with faculty.
- Led the planning and production of high-quality visual content, including WeChat articles, videos, and promotional graphics.

Writing Assistant in Rhetoric 102

Zhejiang, China

Instructor: Mary Lucille Hays

Feb. 2025 – Jun. 2025

- Assisted students in developing academic writing, argumentation, and revision strategies.

Project Leader

Zhejiang, China

Social Practice Project: Exploring Science Communication in the Digital Era

Jan. 2025 – Feb. 2025

- Directed the end-to-end execution of a science communication project, including project planning, outreach, interview coordination, content creation, and publication.
- Interviewed professors, science communicators, and organizations while leading the team in producing high-quality science communication articles and videos.
- Received **Second Prize (Top 3)** among all social practice projects.

Skills

Languages: Chinese (Native), English (TOEFL 5.0/6.0), Cantonese (Basic)

Programming Languages: C++, Python, C, MATLAB, CUDA, Verilog

Frameworks & Tools: PyTorch, ROS 2, Linux, Git